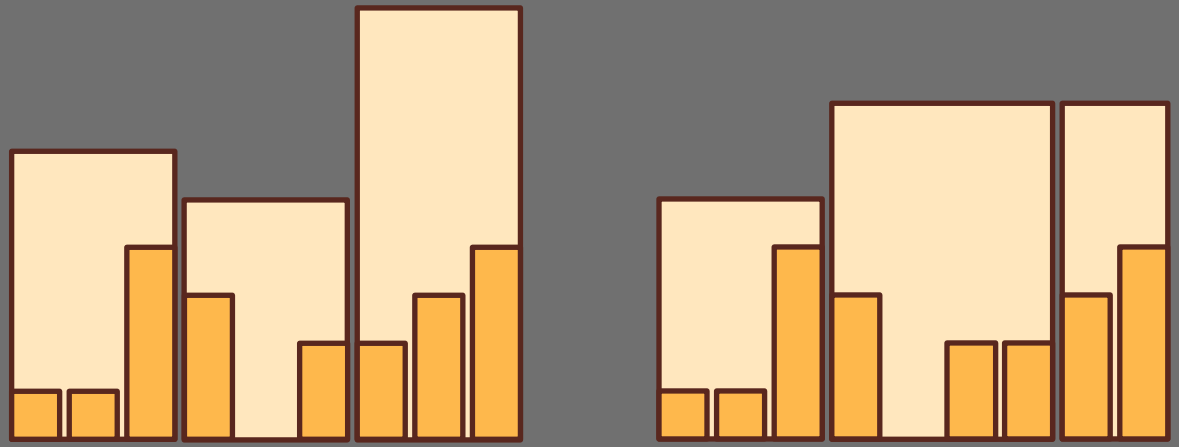
Condition Selectivity Estimation

General principle

- Assume uniform distribution.

- With histogram, assume uniform within block.



Combining selectivities

- $s(A \wedge B) = s(A) \times s(B)$
- $s(\neg A) = 1 - s(A)$
- $s(A \vee B) = 1 - \big(1 - s(A)\big)\big(1 - s(B)\big)$

s(R.a = v)

No histogram

$$s = \frac{1}{values(R.a)}$$

Histogram

Find range $r = [lo, hi]$ containing v.

$$s = \frac{|r|}{|R| \times (hi - lo + 1)}$$

s(R.a = S.b)

View as R.a = v or v’ = S.b.

- Choose the more selective view.
- Assumes each value of attribute with fewer distinct values has a match in the other attribute.

R.a values(R.a)=3	S.b values(S.b)=2	Match?
1	1	Yes
1	2	No
2	1	No
2	2	Yes
3	1	No
3	2	No

I.e., expect 1/3 to match.

s(R.a = S.b)

No histogram

$$s = \min\left(\frac{1}{values(R.a)}, \frac{1}{values(S.b)}\right)$$

$$s = \frac{1}{\max\big(values(R.a), values(S.b)\big)}$$

Histogram

$$s = \min\big(s(R.a = v), s(S.b = v')\big)$$

s(R.a ≥ v)

No histogram

- No statistics: Split data range arbitrarily: $s = \frac{1}{2}$
- Otherwise: Split data range proportionally:
  - If $v \leq maxval(R.a)$,
    $$s = \frac{maxval(R.a) - v}{maxval(R.a) - minval(R.a)}$$
  - Otherwise, $s = 0$

Histogram

$$s = \sum_{v' \geq v} s(R.a = v')$$